EMG-Based Hand Gesture Classification

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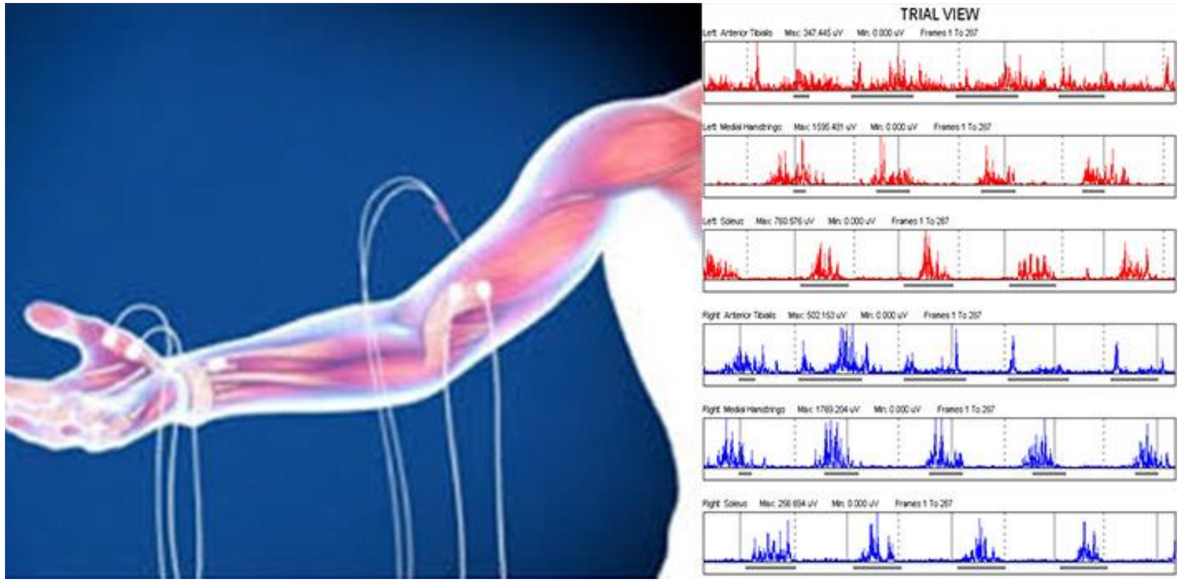
Background

- **Electromyography (EMG)** measures the electrical activity produced by muscle fibers during contraction.

- **Surface EMG (sEMG)** uses non-invasive electrodes placed on the skin to record muscle activation patterns.



Problem & Dataset

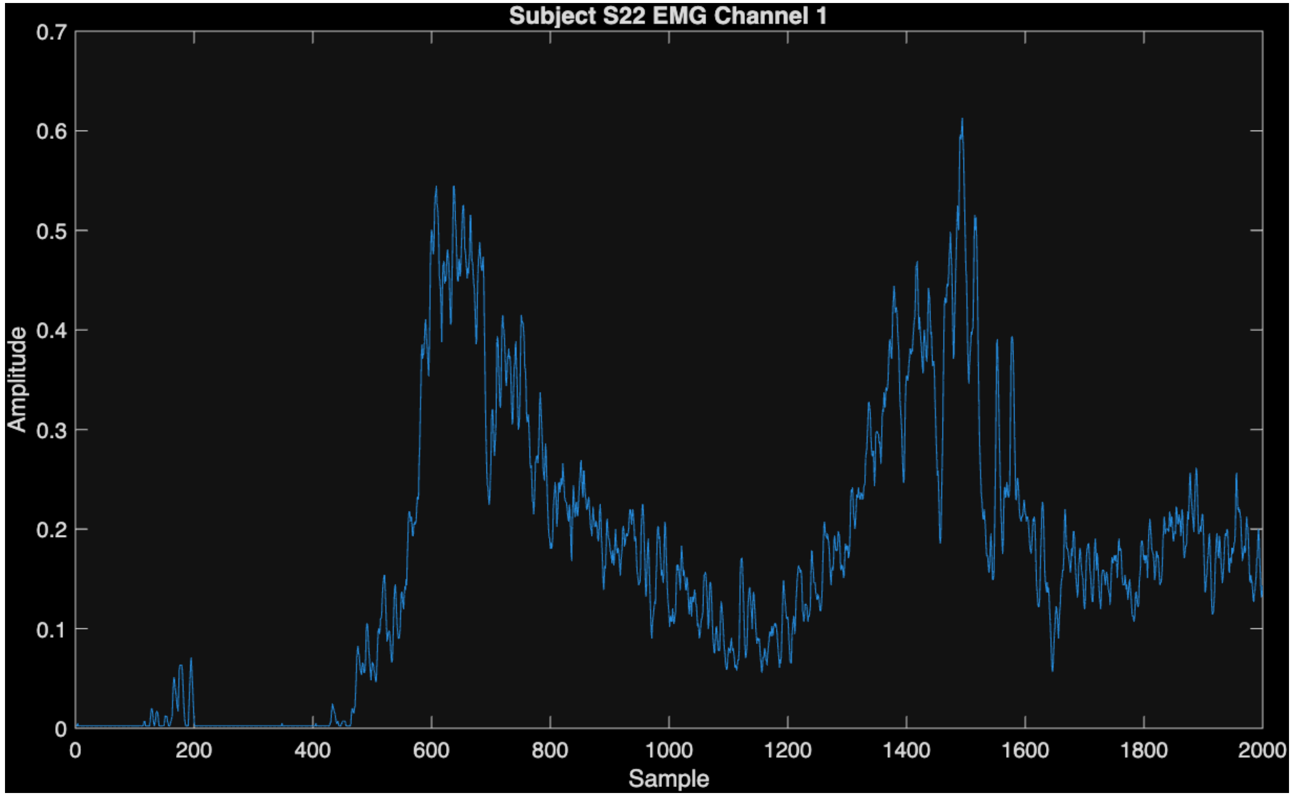
Problem:
Recognize hand gestures using muscle activity measured from 10-channel sEMG.

NinaPro DB1, Exercise A1

23 gestures (tested both 23-class & 12-class subsets)

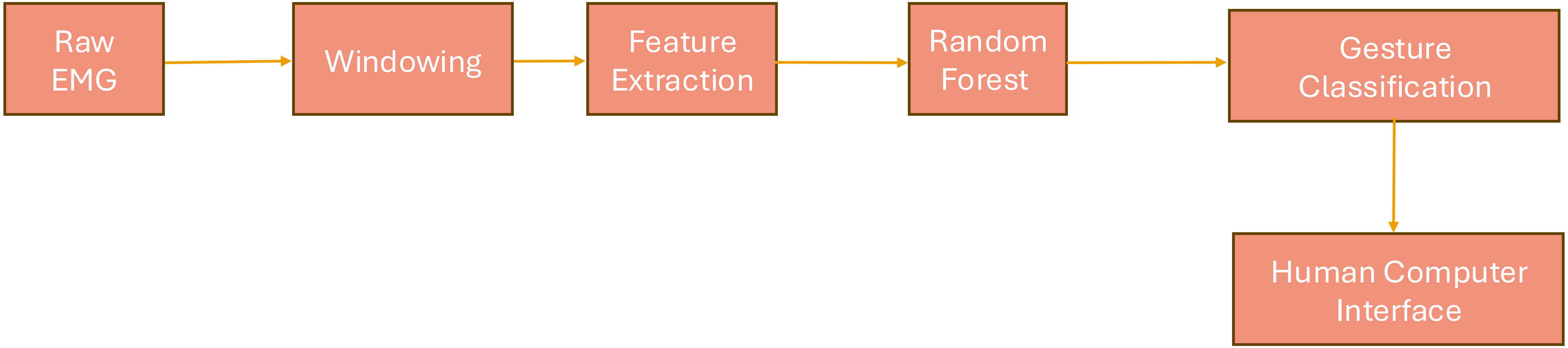
Each sample: 400-sample window × 10 channels

Preprocessed using 50 handcrafted features per window



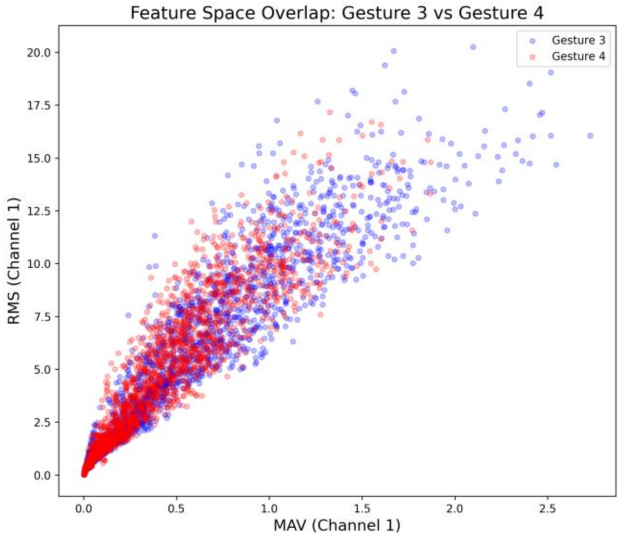
Method- Random Forest Pipeline

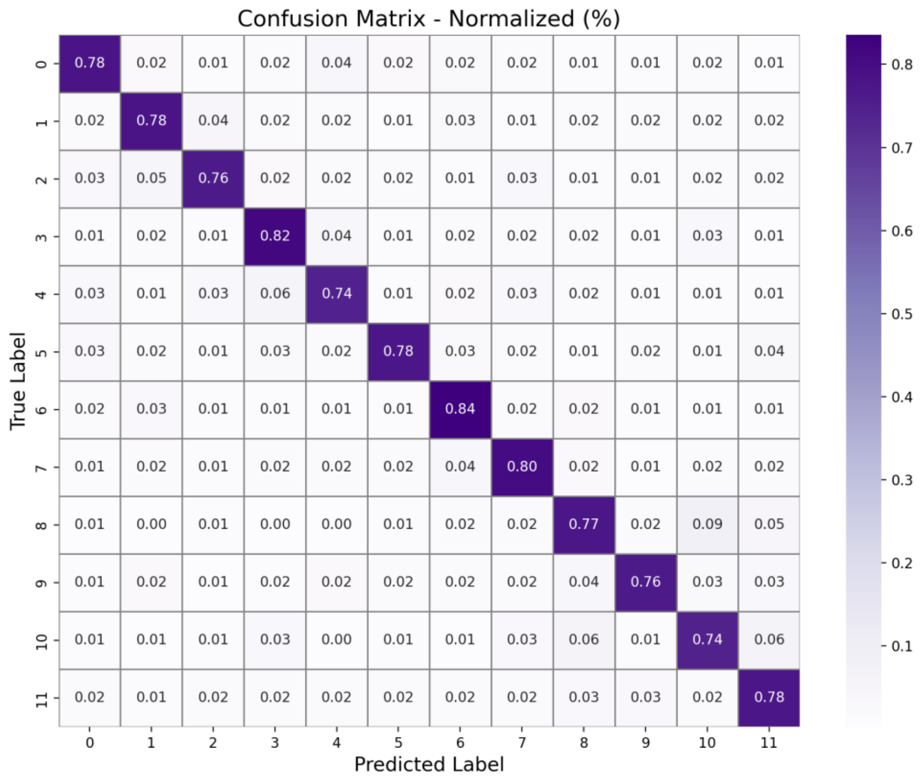
- Windowing (400 samples, 200-ms)
- EMG Features- Mean Absolute Value , Waveform Length, Zero Crossings, Slope Sign Changes, RMS
- Random Forest Classifier 300 trees, max_depth = 30, Train/test = 80/20 split



Experiments & Setup

- **Train/Test Split:** 80/20
- **Classifier:** Random Forest (300 trees)
- **Features:** 50 features (MAV, RMS, WL, ZC, SSC × 10 channels)
- **Accuracy: 78.1%** on test set
- **Avg latency:** ~12.3 ms per prediction





Insight and Limitations

- Features were effective — MAV, RMS, WL, ZC, and SSC captured most EMG variance with 50 dimensions.
- Random Forest achieved 78% accuracy, consistent with published NinaPro baselines.
- Physiological overlap explains gesture confusion — similar gestures activate the same forearm extensor/flexor groups.
- ~12 ms latency, ideal for prosthetic/HCI applications.
- Precision/recall ~0.75–0.83 across all 12 gestures

- Forearm sEMG cannot capture intrinsic hand muscles, limiting separability for gestures.
- High inter-subject variability: electrode placement, skin impedance, and physiology reduce generalization.
- Many gestures rely on overlapping muscle groups, producing similar EMG patterns that limit accuracy.

Future Work

- Implement cross-subject adaptation to improve generalization across different people, electrode placements, and sessions.
- Reduce gesture confusion by applying class-specific feature adaptation for similar gestures.
- Real-time prototype with a microcontroller or prosthetic simulator to test live performance and usability.

Demo

```
Last login: Wed Dec  3 23:38:35 on ttys005
(base) ashwinsakthivel@Ashwins-MacBook-Air-3 ~ % cd Documents
(base) ashwinsakthivel@Ashwins-MacBook-Air-3 Documents % cd projectemg
(base) ashwinsakthivel@Ashwins-MacBook-Air-3 projectemg % cd EMG-Based-Hand-Recognition
(base) ashwinsakthivel@Ashwins-MacBook-Air-3 EMG-Based-Hand-Recognition % streamlit run UI/emg_ui.py
```

Thank You